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GENERAL INSTRUCTIONS

DESCRIPTION: HELICAL COIL INSERTS ARE SCREW THREAD BUSHINGS COILED FROM DIAMOND-SHAPED CROSS-SECTION WIRE. THEY ARE SCREWED INTO TAPPED HOLES TO FORM NOMINAL SIZE INTERNAL THREADS. INSERTS ARE INSTALLED BY TORQUING THROUGH A DIAMETRAL TANG WHICH IS NOTCHED FOR TANG REMOVAL.

THREADS: ASSEMBLED INSERT, CLASSES 2B AND 3B ARE CONTROLLED BY TOLERANCE RANGE OF TAPPED HOLE INTO WHICH INSERT IS FITTED. THEREFORE, THE SAME INSERT IS USED FOR THREAD CLASSES 2B AND 3B. THE GRIP COIL OR COILS OF THE SCREW LOCKING INSERT ARE SHAPED TO PROVIDE PREVAILING TORQUE.

TAPPED HOLE: 60° AMERICAN NATIONAL OR UNIFIED FORM. TAPS AND GAGES STANDARD WITH MANUFACTURER.

NOMINAL LENGTH OF INSERT: NOMINAL INSERT LENGTHS ARE EQUAL TO 1, 1-1/2 AND 2 TIMES THE NOMINAL MAJOR DIAMETER OF THE SCREW THREAD FOR FREE RUNNING INSERTS AND 1-1/2 AND 2 TIMES THE NOMINAL MAJOR DIAMETER OF THE SCREW THREAD FOR SCREW LOCKING INSERTS. FOR AVAILABLE NOMINAL LENGTHS OF SPECIFIC INSERTS, SEE STANDARD NSA21209 (INSERTS, SCREW THREAD, COARSE AND FINE, SCREW LOCK, HELICAL COIL) AND NSA21206 (INSERTS, SCREW THREAD, COARSE AND FINE, FREE RUNNING, HELICAL COIL).

LENGTH OF ASSEMBLED INSERT: THE MINIMUM LENGTH, L, OF ASSEMBLED INSERT IS EQUAL TO THE NOMINAL LENGTH MINUS 3/4 PITCH. MAXIMUM LENGTH OF ASSEMBLED INSERT IS EQUAL TO MINIMUM LENGTH, L, PLUS 1/4 PITCH.

DEPTH OF RECOMMENDED MINIMUM TAP DRILLED HOLE: FOR BLIND HOLES WITH A COUNTERSINK, AS SPECIFIED IN THE TABULATION, THE MINIMUM TAP DRILL DEPTH, F, EQUALS THE MAXIMUM LENGTH OF ASSEMBLED INSERT PLUS 6 PITCHES WHEN STANDARD PLUG TAPS ARE USED. FOR BLIND HOLES WITHOUT COUNTERSINK THE MINIMUM TAP DRILL DEPTH FOR STANDARD PLUG TAPS IS EQUAL TO MAXIMUM LENGTH OF ASSEMBLED INSERT PLUS 5 PITCHES. WHEN DESIGN WILL NOT PERMIT THE ABOVE TAP DRILL DEPTHS FOR STANDARD PLUG TAPS, BOTTOMING TAPS MAY BE USED WHICH WILL ALLOW THE TAP DRILL DEPTH TO BE DECREASED UP TO 3 PITCHES. IF TAP DRILL DEPTH IS DECREASED BY THE MAXIMUM OF 3 PITCHES, NO TAP END CLEARANCE IS PROVIDED.

DEPTH OF FULL THREAD OF TAPPED HOLE: FOR THRU OR BLIND TAPPED HOLES WITH A COUNTERSINK, AS SPECIFIED IN THE TABULATION, THE MINIMUM FULL THREAD, H (ALSO MINIMUM FLANGE THICKNESS FOR THRU HOLE), EQUALS MINIMUM ASSEMBLED LENGTH OF INSERT, L, AS SPECIFIED IN THE TABULATION PLUS 1-3/4 PITCH. FOR THRU OR BLIND TAPPED HOLES WITHOUT COUNTERSINK, THE MINIMUM FULL THREAD (ALSO MINIMUM FLANGE THICKNESS FOR THRU HOLE) SHALL NOT BE LESS THAN THE MINIMUM LENGTH OF INSERT, L, PLUS 3/4 PITCH.

LENGTH OF BOLT PROJECTION (WITH COUNTERSINK HOLE): THE MAXIMUM LENGTH OF BOLT THREAD PROJECTION, K, AS SPECIFIED IN THE TABULATION (WHEN INSERT TANG IS NOT REMOVED), IS BASED ON THE MINIMUM LENGTH OF FREE RUNNING (CLASS 1) INSERT PLUS 1/4 PITCH. WHEN TANG IS REMOVED, THE SPECIFIED VALUES FOR K MAY BE INCREASED TO EQUAL THE MINIMUM DESIGN DEPTH OF THE TAP DRILL HOLE, F. K DIMENSIONS ARE NOT APPLICABLE TO SCREW-LOCK INSERTS FOR THE SIZES LISTED BELOW. FOR THESE SIZES, TANG SHOULD BE REMOVED TO PERMIT ENGAGEMENT OF SCREW WITH GRIP COIL IN ACCORDANCE WITH REQUIREMENTS OF K.

1-1/2 DIAMETER LONG UNC 4-40 THRU 7/16-14
2 DIAMETER LONG UNC 4-40 THRU 1/4-20
1-1/2 DIAMETER LONG UNF 10-32 THRU 5/16-24
2 DIAMETER LONG UNF 10-32 AND SMALLER.

MINIMUM FULL THREAD PROJECTION: DIMENSION H IS MINIMUM DISTANCE THAT FIRST FULL THREAD OF BOLT MUST PROJECT BELOW WORK SURFACE TO INSURE A LOCKING EFFECT FOR SCREW-LOCKING INSERTS (LOCATED 1-1/2 P MAX. BELOW SURFACE OF WORK).

LENGTH OF BOLT PROJECTION (WITHOUT COUNTERSINK HOLE): FOR TAPPED HOLES, WITHOUT COUNTERSINK AND HAVING INSERTS INSTALLED AT A MINIMUM OF 1/4 PITCH BELOW THE BOSS SURFACE, THE SPECIFIED VALUES FOR K SHOULD BE DECREASED 3/4 PITCH WHEN TANG IS NOT REMOVED.

COUNTERSINK AND COUNTERBORE: THE VALUES GIVEN IN THE TABULATION FOR DEPTH OF HOLE, F; MINIMUM FULL THREAD, H; AND LENGTH OF BOLT PROJECTION, K, ARE MEASURED FROM THE SURFACE OF THE BOSS OR FIRCE AND ARE BASED ON THE COUNTERSINK ANGLE TO H DIAMETER SHOWN. IF A COUNTERBORE OR COUNTERSINK OTHER THAN THAT SHOWN IS REQUIRED, THE VALUES FOR F, H AND K MUST BE MODIFIED.

INSTALLATION OF INSERT (WITH COUNTERSINK HOLE): THE TOP EDGE OF THE INSERT SHALL BE INSTALLED 3/4 PITCH TO 1-1/2 PITCHES BELOW TOP SURFACE OF TAPPED HOLE.

BLIND HOLE ASSEMBLY (WITHOUT REMOVAL OF TANG): TO OBTAIN DESIGN BOLT-INSERT THREAD ENGAGEMENT EQUIVALENT TO A NOMINAL INSERT LENGTH WITHOUT REMOVING THE INSERT TANG, AN INSERT 1/2 DIAMETER GREATER IN NOMINAL LENGTH SHALL BE USED.

PERMITTED MODIFICATIONS: VALUES FOR F, H, K, L, AND INSTALLATION DEPTH OF INSERT MAY BE MODIFIED TO SUIT REQUIREMENTS FOR PRODUCTION TOOLING, DESIGN, ETC.

INSTALLATION OF INSERT (WITHOUT COUNTERSINK HOLE): FOR MINIMUM FLANGE THICKNESS (EQUAL TO NOMINAL INSERT LENGTH), THE TOP EDGE OF THE INSERT SHALL BE INSTALLED 1/4 PITCH TO 1/2 PITCH BELOW TOP SURFACE OF TAPPED HOLE.

Ⓐ CANCELED AFTER 11 JANUARY 1971. USE MS33537.

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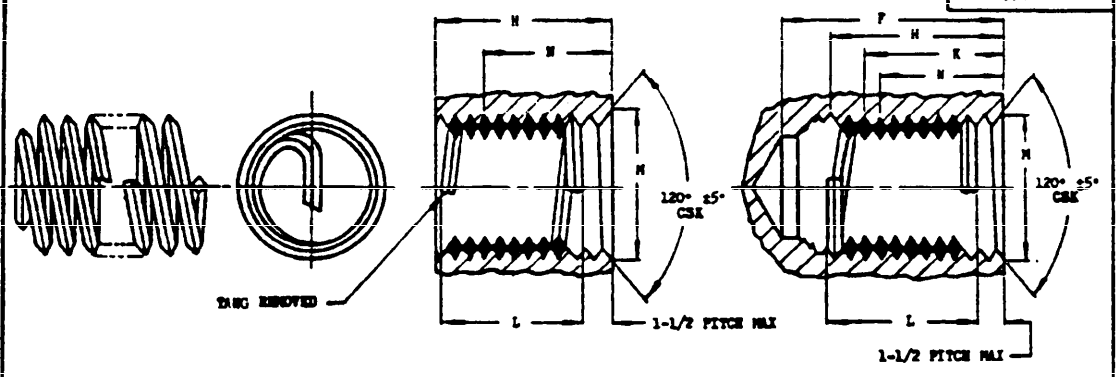
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P.A. USAF - AFSC Other Cust Army - CE Navy - Navy	TITLE INSERT, SCREW THREADS, HELICAL COIL, ASSEMBLY AND HOLE, DIMENSIONS FOR	MILITARY STANDARD MS33646
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 1 OF 5

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	NOMINAL SIZE	.112	.138	.164	.190		.250		.3125	
	THREADS PER INCH	40	32	32	24 (a)	32	20 (a)	28	18 (a)	24
ASSEMBLED INSERT	PD, MAX, CLASS 2B	.0991	.1214	.1475	.1672	.1736	.2223	.2311	.2817	.2902
	PD, MAX, CLASS 3B	.0982	.1204	.1465	.1661	.1726	.2211	.2300	.2803	.2890
	PD, MIN, ALL CLASSES	.0958	.1177	.1437	.1629	.1697	.2175	.2268	.2764	.2854
	MINOR DIA, MAX, CLASS 2B	.0939	.1140	.139	.156	.164	.207	.220	.265	.277
	MINOR DIA, MAX, CLASS 3B	.0939	.1140	.1389	.1555	.1641	.2067	.2190	.2630	.2754
	MINOR DIA, MIN, ALL CLASSES	.0849	.1040	.1300	.1450	.1560	.1960	.2110	.2520	.2670
	L MIN LENGTH OF INSERT WHEN									
	NOMINAL LENGTH = 1 DIA	.093	.115	.141	.159	.167	.212	.223	.271	.281
	NOMINAL LENGTH = 1-1/2 DIA	.149	.184	.223	.254	.262	.338	.348	.427	.438
	NOMINAL LENGTH = 2 DIA	.205	.253	.305	.349	.357	.462	.473	.583	.594
TAPPED HOLE	F DEPTH OF BLIND HOLE, MIN WHEN									
	NOMINAL LENGTH = 1 DIA	.250	.310	.336	.419	.362	.525	.486	.617	.581
	NOMINAL LENGTH = 1-1/2 DIA	.306	.379	.418	.514	.457	.650	.571	.773	.697
	NOMINAL LENGTH = 2 DIA	.362	.448	.500	.609	.552	.775	.696	.930	.854
	H MIN FULL THREAD WHEN									
	NOMINAL LENGTH = 1 DIA	.140	.170	.200	.230	.220	.300	.290	.370	.360
	NOMINAL LENGTH = 1-1/2 DIA	.190	.240	.280	.330	.320	.430	.410	.530	.510
	NOMINAL LENGTH = 2 DIA	.250	.310	.360	.420	.410	.550	.540	.680	.670
	PD, MAX FOR									
	CLASS 2B INSERTS	.1306	.1611	.1872	.2203	.2133	.2863	.2765	.3529	.3433
	CLASS 3B INSERTS	.1299	.1601	.1862	.2192	.2123	.2851	.2754	.3515	.3421
	PD, MIN, ALL CLASSES	.1283	.1583	.1843	.2170	.2103	.2825	.2732	.3486	.3395
	MINOR DIA, MAX, ALL CLASSES	.121	.150	.176	.205	.202	.267	.264	.334	.330
	MINOR DIA, MIN, ALL CLASSES	.116	.144	.170	.199	.196	.261	.257	.328	.323
	MAJOR DIA, MIN, ALL CLASSES	.1404	.1733	.1993	.2370	.2253	.3062	.2904	.3751	.3595
	K MAX LENGTH OF BOLT PROJECTION IF TANG IS NOT REMOVED, WHEN INSERT									
	NOMINAL LENGTH = 1 DIA	.100	.122	.149	.169	.175	.225	.232	.285	.291
	NOMINAL LENGTH = 1-1/2 DIA	.156	.191	.233	.264	.270	.351	.357	.443	.448
	NOMINAL LENGTH = 2 DIA	.212	.260	.313	.359	.365	.475	.482	.597	.604
	N COUNTERSINK DIAMETER									
	MAXIMUM	.180	.210	.240	.280	.260	.340	.330	.410	.400
	MINIMUM	.140	.170	.200	.240	.220	.330	.290	.370	.360
	M MIN LENGTH OF BOLT FIRST FULL THREAD PROJECTION FOR THE SCREW LOCKING INSERT ASSEMBLY, WHEN INSERT									
	NOMINAL LENGTH = 1-1/2 DIA	.16	.19	.21	.26	.23	.33	.29	.40	.38
	NOMINAL LENGTH = 2 DIA	.19	.23	.26	.31	.28	.40	.36	.48	.46

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PROCUREMENT SPECIFICATION	SUPERSEDES	SHEET 2

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NOMINAL SIZE			.375		.4375		.500		.5625	
THREAD PER INCH			16 (a)	24	14 (a)	20	13 (a)	20	12 (a)	18
ASSEMBLED INSERT		PD, MAX, CLASS 2B	.3401	.3528	.3972	.4104	.4565	.4731	.5152	.5323
		PD, MAX, CLASS 3B	.3387	.3516	.3957	.4091	.4548	.4717	.5135	.5308
		PD, MIN, ALL CLASSES	.3344	.3479	.3911	.4050	.4500	.4675	.5084	.5264
		MINOR DIA, MAX, CLASS 2B	.321	.340	.376	.395	.434	.457	.490	.515
		MINOR DIA, MAX, CLASS 3B	.3182	.3372	.3717	.3916	.4294	.4537	.4843	.5106
		MINOR DIA, MIN, ALL CLASSES	.3070	.3300	.3600	.3830	.4170	.4460	.4720	.5020
	L	MIN LENGTH OF INSERT WHEN								
		NOMINAL LENGTH = 1 DIA	.328	.344	.384	.400	.442	.462	.500	.521
		NOMINAL LENGTH = 1-1/2 DIA	.516	.531	.603	.619	.692	.712	.781	.808
		NOMINAL LENGTH = 2 DIA	.703	.719	.821	.838	.942	.962	1.062	1.083
TAPPED HOLE	P	DEPTH OF BLIND HOLE, MIN WHEN								
		NOMINAL LENGTH = 1 DIA	.719	.604	.830	.712	.924	.775	1.021	.867
		NOMINAL LENGTH = 1-1/2 DIA	.906	.791	1.069	.931	1.174	1.025	1.302	1.148
		NOMINAL LENGTH = 2 DIA	1.093	.979	1.267	1.150	1.424	1.275	1.583	1.430
	H	MIN FULL THREAD WHEN								
		NOMINAL LENGTH = 1 DIA	.440	.420	.510	.490	.580	.550	.650	.680
		NOMINAL LENGTH = 1-1/2 DIA	.630	.600	.730	.710	.830	.800	.930	.900
		NOMINAL LENGTH = 2 DIA	.810	.790	.950	.930	1.080	1.050	1.210	1.180
		PD, MAX, FOR								
		CLASS 2B INSERTS	.4203	.4059	.4890	.4744	.5554	.5371	.6225	.6035
		CLASS 3B INSERTS	.4189	.4047	.4875	.4731	.5537	.5357	.6208	.6020
		PD, MIN, ALL CLASSES	.4156	.4020	.4839	.4700	.5499	.5325	.6167	.5986
		MINOR DIA, MAX, ALL CLASSES	.398	.392	.463	.458	.525	.522	.588	.586
		MINOR DIA, MIN, ALL CLASSES	.390	.385	.453	.450	.515	.513	.578	.577
		MAJOR DIA, MIN, ALL CLASSES	.4454	.4220	.5179	.4937	.5865	.5562	.6562	.6251
	K	MAX LENGTH OF BOLT PROJECTION IF TANG IS NOT REMOVED, WHEN INSERT								
		NOMINAL LENGTH = 1 DIA	.344	.354	.402	.413	.461	.475	.521	.535
		NOMINAL LENGTH = 1-1/2 DIA	.532	.541	.621	.632	.711	.725	.802	.816
		NOMINAL LENGTH = 2 DIA	.719	.729	.840	.851	.961	.975	1.083	1.097
	M	COUNTERSINK DIAMETER								
		MAXIMUM	.480	.460	.560	.530	.620	.590	.690	.660
		MINIMUM	.440	.420	.520	.490	.580	.550	.650	.620
	N	MIN LENGTH OF BOLT FIRST FULL THREAD PROJECTION FOR THE SCREW LOCKING INSERT ASSEMBLY, WHEN								
		NOMINAL LENGTH = 1-1/2 DIA	.46	.42	.54	.50	.60	.55	.67	.61
		NOMINAL LENGTH = 2 DIA	.56	.52	.65	.61	.73	.67	.81	.75

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P.A. USAF - AFSC Other Cust Army - CE Navy - aop	TITLE INSERT, SCREW THREADS, HELICAL COIL, ASSEMBLY AND HOLE, DIMENSIONS FOR	MILITARY STANDARD MS33646
PROCUREMENT SPECIFICATION	SUPERSEDES	SHEET 3

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NOMINAL SIZE			.625		.750		.875		1.0000	
THREAD PER INCH			11 (a)	18	10 (a)	16	9 (a)	14	8 (a)	12
ASSEMBLED INSERT		PD, MAX, CLASS 2B	.5732	.5969	.6927	.7159	.8110	.8356	.9276	.9535
		PD, MAX, CLASS 3B	.5714	.5934	.6907	.7143	.8089	.8339	.9254	.9516
		PD, MIN, ALL CLASSES	.5660	.5889	.6850	.7094	.8028	.8286	.9188	.9459
		MINOR DIA, MAX, CLASS 2B	.546	.578	.663	.696	.778	.814	.890	.928
		MINOR DIA, MAX, CLASS 3B	.5391	.5730	.6545	.6908	.7681	.8068	.8797	.919*
		MINOR DIA, MIN, ALL CLASSES	.5270	.5650	.6420	.6820	.7550	.7980	.8650	.9100
	L	MIN LENGTH OF INSERT WHEN NOMINAL LENGTH = 1 DIA	.557	.583	.675	.703	.792	.821	.906	.938
		NOMINAL LENGTH = 1-1/2 DIA	.869	.896	1.050	1.078	1.229	1.259	1.406	1.438
		NOMINAL LENGTH = 2 DIA	1.182	1.208	1.425	1.453	1.667	1.696	1.906	1.938
TAPPED HOLE	F	DEPTH OF BLIND HOLE, MIN WHEN NOMINAL LENGTH = 1 DIA	1.125	.930	1.300	1.094	1.486	1.267	1.688	1.583
		NOMINAL LENGTH = 1-1/2 DIA	1.437	1.242	1.675	1.469	1.924	1.704	2.188	2.083
		NOMINAL LENGTH = 2 DIA	1.750	1.555	2.050	1.844	2.361	2.142	2.688	2.581
	H	MIN FULL THREAD WHEN NOMINAL LENGTH = 1 DIA	.720	.680	.950	.810	.990	.950	1.130	1.125
		NOMINAL LENGTH = 1-1/2 DIA	1.030	.990	1.230	1.190	1.420	1.380	1.630	1.625
		NOMINAL LENGTH = 2 DIA	1.340	1.310	1.600	1.560	1.860	1.820	2.130	2.125
		PD, MAX FOR CLASS 2B INSERTS	.6903	.6661	.8216	.7961	.9543	.9278	1.0890	1.0608
		CLASS 3B INSERTS	.6885	.6646	.8196	.7945	.9522	.9257	1.0868	1.0589
		PD, MIN, ALL CLASSES	.6841	.6611	.8149	.7906	.9471	.9214	1.0812	1.0542
		MINOR DIA, MAX, ALL CLASSES	.663	.649	.791	.775	.916	.900	1.044	1.025
		MINOR DIA, MIN, ALL CLASSES	.653	.640	.781	.765	.906	.890	1.031	1.015
		MAJOR DIA, MIN, ALL CLASSES	.7272	.6876	.9623	.9204	.9996	.9554	1.1402	1.0938
	K	MAX LENGTH OF BOLT PROJECTION IF TANG IS NOT REMOVED, WHEN INSERT NOMINAL LENGTH = 1 DIA	.580	.597	.700	.719	.820	.840	.937	.958
		NOMINAL LENGTH = 1-1/2 DIA	.892	.910	1.075	1.094	1.257	1.277	1.437	1.458
		NOMINAL LENGTH = 2 DIA	1.205	1.222	1.450	1.469	1.695	1.714	1.937	1.958
	N	COUNTERSINK DIAMETER MAXIMUM	.760	.720	.900	.860	1.040	.990	1.180	1.130
		MINIMUM	.720	.680	.960	.820	1.000	.950	1.140	1.110
	M	MIN LENGTH OF BOLT FIRST FULL THREAD PROJECTION FOR THE SCREW LOCKING INSERT ASSEMBLY, WHEN INSERT NOMINAL LENGTH = 1-1/2 DIA	.74	.66	.86	.78	.98	.90	1.12	1.04
		NOMINAL LENGTH = 2 DIA	.89	.82	1.04	.97	1.20	1.12	1.37	1.29

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P.A. USAF - AFSC

Other Cust
Army - CE
Navy - Nsp

TITLE

INSERT, SCREW THREADS, HELICAL COIL, ASSEMBLY AND HOLE, DIMENSIONS FOR

MILITARY STANDARD

MS33646

PROCUREMENT SPECIFICATION

SUPERSEDES

SHEET 4

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NOMINAL SIZE			1.125		1.250		1.375		1.500	
THREADS PER INCH			7 (a)	12	7 (a)	12	6 (a)	12	6 (a)	12
ASSEMBLED INSERT		FD, MAX, CLASS 2B	1.0416	1.0787	1.1668	1.2039	1.2771	1.3291	1.4022	1.4542
		FD, MAX, CLASS 3B	1.0393	1.0768	1.1644	1.2019	1.2745	1.3270	1.3996	1.4522
		FD, MIN, ALL CLASSES	1.0322	1.0709	1.1572	1.1959	1.2667	1.3209	1.3917	1.4459
		MINOR DIA, MAX, CLASS 2B	.998	1.053	1.123	1.172	1.225	1.303	1.350	1.428
		MINOR DIA, MAX, CLASS 3B	.9875	1.0443	1.1125	1.1698	1.2146	1.2948	1.3396	1.4198
		MINOR DIA, MIN, ALL CLASSES	.9703	1.0350	1.0950	1.1600	1.1950	1.2850	1.3200	1.4100
	L	MIN LENGTH OF INSERT WHEN NOMINAL LENGTH = 1 DIA	1.018	1.062	1.143	1.188	1.250	1.312	1.375	1.438
		NOMINAL LENGTH = 1-1/2 DIA	1.580	1.625	1.768	1.812	1.938	2.000	2.125	2.188
		NOMINAL LENGTH = 2 DIA	2.143	2.188	2.393	2.438	2.625	2.688	2.875	2.938
TAPPED HOLE	F	DEPTH OF BLIND HOLE, MIN WHEN NOMINAL LENGTH = 1 DIA	1.911	1.583	2.036	1.708	2.292	1.833	2.417	1.958
		NOMINAL LENGTH = 1-1/2 DIA	2.473	2.146	2.661	2.333	2.979	2.521	3.167	2.708
		NOMINAL LENGTH = 2 DIA	3.036	2.708	3.286	2.958	3.667	3.208	3.917	3.458
	W	MIN FULL THREAD WHEN NOMINAL LENGTH = 1 DIA	1.270	1.210	1.390	1.330	1.510	1.460	1.670	1.580
		NOMINAL LENGTH = 1-1/2 DIA	1.830	1.770	2.020	1.960	2.230	2.150	2.420	2.330
		NOMINAL LENGTH = 2 DIA	2.390	2.330	2.640	2.580	2.920	2.830	3.170	3.080
		FD, MAX FOR CLASS 2B INSERTS	1.2262	1.1860	1.3514	1.3112	1.4926	1.4364	1.6177	1.5615
		CLASS 3B INSERTS	1.2239	1.1841	1.3490	1.3092	1.4900	1.4343	1.6151	1.5595
		FD, MIN, ALL CLASSES	1.2178	1.1792	1.3428	1.3042	1.4832	1.4292	1.6082	1.5542
		MINOR DIA, MAX, ALL CLASSES	1.186	1.159	1.311	1.284	1.436	1.409	1.571	1.534
		MINOR DIA, MIN, ALL CLASSES	1.171	1.140	1.296	1.265	1.421	1.390	1.546	1.515
		MAJOR DIA, MIN, ALL CLASSES	1.2850	1.2187	1.4100	1.3437	1.5620	1.4687	1.6870	1.5937
	K	MAX LENGTH OF BOLT PROJECTION IF TANG IS NOT REMOVED, WHEN INSERT NOMINAL LENGTH = 1 DIA	1.054	1.083	1.179	1.208	1.292	1.333	1.417	1.458
		NOMINAL LENGTH = 1-1/2 DIA	1.616	1.646	1.804	1.833	1.979	2.021	2.167	2.208
		NOMINAL LENGTH = 2 DIA	2.179	2.208	2.429	2.458	2.667	2.708	2.917	2.958
	N	COUNTERSINK DIAMETER MAXIMUM	1.320	1.260	1.450	1.380	1.600	1.510	1.730	1.630
		MINIMUM	1.280	1.220	1.410	1.340	1.560	1.470	1.690	1.590
	N	MIN LENGTH OF BOLT FIRST FULL THREAD PROJECTION FOR THE SCREW LOCKING INSERT ASSEMBLY, WHEN INSERT NOMINAL LENGTH = 1-1/2 DIA								
		NOMINAL LENGTH = 2 DIA								

(a) NOT RECOMMENDED FOR NEW DESIGN, SEE MIL-S-7742 FOR RECOMMENDED SERIES.

THIS IS A DESIGN STANDARD. NOT TO BE USED AS A PART NUMBER.

APPROVED 17 Jul 81 REVISED (A) FOR CHANGES SEE SHEET 1.

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P.A. USAF - AFSC Other Cust Army - CE Navy - SHIP	TITLE INSERTS, SCREW THREADS, HELICAL COIL, ASSEMBLY AND HOLES, DIMENSIONS FOR	MILITARY STANDARD MS33646
PROCUREMENT SPECIFICATION	SUPERSEDES	SHEET 5

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